Minimizing Environmental Magnetic Field Sources for nEDM

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ABSTRACT

Measurement of the neutron's Electric Dipole Moment (nEDM) could potentially explain why there is more matter than antimatter in the universe, and it would suggest plausible extensions to the Standard Model. We will attempt to detect the nEDM by measuring the electric-field-dependent neutron precession frequency, which is highly sensitive to magnetic field gradients. In order to produce fields with sufficiently low gradients for our experiment, we eliminate environmental effects by offsetting the ambient field with a Field Compensation System (FCS), then magnetically shielding the reduced field with a Mu-Metal cylinder. We discovered that the strongest environmental effect in our lab came from iron rebar embedded in the floor beneath the proposed experiment location. The large extent and strength of the floor's magnetization made the effect too large to offset with the FCS, forcing us to relocate our apparatus. The floor's magnetic field was mapped with a Hall probe in order to determine the most viable experiment locations. A 3-axis Fluxgate magnetometer was then used to determine the floor field's drop-off and shape at these locations, and a final apparatus position was determined which minimized the floor's effect such that it could be effectively offset and shielded by our experiment.

1. Introduction

Though a neutron is a net neutral particle, it should theoretically have an electric dipole moment (EDM), arising from the elementary quark and quark-chromo EDMs of its constituent elementary particles (one +2e/3 u-quark, and two –e/3 d-quarks)[1]. Though the neutron electric dipole moment (nEDM) is very difficult to detect because it's very small, measurement of the nEDM at a magnitude larger than that predicted by the Standard Model ($10^{-31}\mathrm{e}\cdot\mathrm{cm}$) could have profound consequence[2] on theoretical physics, because it would suggest viable extensions for particle physics beyond the Standard Model, and it may also yield an explanation to the Baryon Asymmetry Problem in the form of increased Charge-Parity symmetry violation – one of the *Sakharov Conditions*[3] necessary for Baryogenesis. With an expected sensitivity of $2*10^{-28}\mathrm{e}\cdot\mathrm{cm}$ [4], nEDM@SNS will search for the neutron's electric dipole moment with unprecedented precision, capable of detecting an nEDM a full 2 orders of magnitude smaller than the current experimental upper bound[4,5].

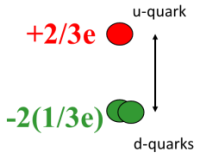


Figure 1[4]: The neutron EDM could arise from a nonzero average separation between quarks.

In order to identify the nEDM, our experiment will attempt to measure frequency shifts in neutron spin precession as a result of changing the magnitude of the electric field. As shown by the Larmor precession frequency equation[6], the neutron's spin precession frequency should be linearly dependent on E, scaling as $\pm\frac{2d_n}{\hbar}$:

- Larmor Frequency:

$$\omega_n = -2\frac{\mu_n B_0 \pm d_n E}{\hbar}$$

Where:
ω_n = precession frequency, μ_n = magnetic dipole moment, d_n = electric dipole moment,
B_0 = strength of magnetic holding field, E = strength of applied electric field, $\hbar$ = reduced Planck's constant

In the apparatus, we'll trap *ultracold neutrons (UCN)* in a uniform magnetic *holding field* [7]. The Larmor precession frequency of the UCNs can then be observed from the rate of scintillation[7], caused by the interaction of the UCN dipoles with the helium atoms that spin-dress them.

Unfortunately, non-zero magnetic field gradients will introduce additional neutron spin precession frequency shift, which is also linear with E. This additional frequency shift is known as the geometric phase effect [8]:

- Systematic Uncertainty from Geometric Phase Effect:

$$\delta\omega_n(\omega_0) = -\gamma^2\frac{E}{c}(\omega_0\mathrm{Im}[G_y S_{yy}(\omega_0) + G_z S_{zz}(\omega_0)] + G_y\frac{{L_y}^2}{12} + G_z\frac{{L_z}^2}{12})$$

Where:
$\delta\omega_n$= frequency shift, γ = neutron gyromagnetic ratio, ω_0 = precession frequency due to holding field only,
c = speed of light, S_{ii} = position correlation function spectrum, L_j = length of measurement cell in j direction,
G_k = linear magnetic field gradient wrt k

This geometric phase effect will add a systematic uncertainty to our results. In the most recent nEDM experiment (Sussex/RAL/ILL), "geometric phase" was the largest source of systematic uncertainty[8]. Minimizing the spatial gradient of our apparatus' holding field is essential for our experiment to achieve its desired level of sensitivity.

Gradient minimization of the holding field is carried out by layers of magnetic shielding that enclose the measurement cell. Most of the layers are composed of materials with high magnetic permeabilities (Mu-Metal, Metglas), allowing us to divert external field lines away from their enclosed volumes.

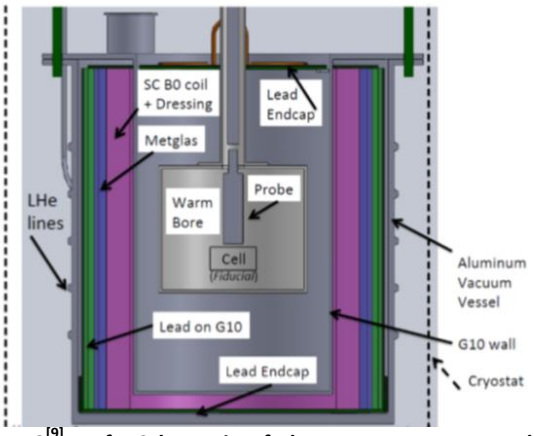


Figure 2[9]: Left: Schematic of the nEDM apparatus. The precession frequency is measured by the fiducial cell, and the B-field is measured by the probe.

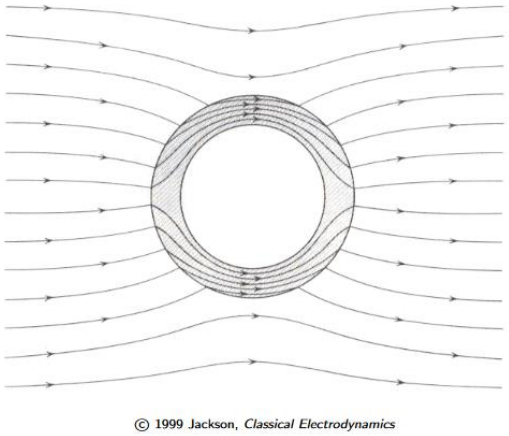


Figure 3[10]: A cylinder of high magnetic permeability diverts field lines away from the volume that it encloses.

Magnetic shielding is expensive, so the entire apparatus will be surrounded by a **Field Compensation System (FCS)**, a prism of rectangular current loops designed to offset large ambient fields. The FCS allows us to use thinner layers of magnetic shielding and still avoid saturation, reducing material costs. The FCS was originally proposed as a way to offset the earth's magnetic field, but our group discovered that this wasn't the only large environmental factor in lab: The floor was also heavily magnetized.



Figure 4: Third-scale apparatus. The outermost magnetic shielding cylinder (centered) is surrounded by the FCS.

In this paper, I will first discuss the discovery and subsequent analysis of the floor magnetization. I will then describe an FCS simulation that I created, and explain how I was able to use this in conjunction with my floor analysis in order to effectively mitigate problems arising due to the floor's magnetic field.

2. Floor Magnetization

2.1. Expected Field

The earth's magnetic field can be likened to that of a massive dipole, so the local field can be estimated based on the location's radial distance from the center of the earth, and its azimuthal angle from magnetic north. The estimated Earth Field in Pasadena is shown in the figure below:

Approximate Earth Field as a Giant Dipole:

$$B_r = -2\,B_0\left(\frac{R_E}{r}\right)^3 \mathrm{Cos}(\theta)$$

$$B_\theta = -B_0\left(\frac{R_E}{r}\right)^3 \mathrm{Sin}(\theta)$$

Using $B_0 \approx 0.312$ G, $\theta \approx 55^\circ$, $r \approx R_E$

$$B_r \approx 0.36\ G$$
$$B_\theta \approx 0.26\ G$$

B-field measured in office:

$$B_x \approx 0.06\ G$$
$$B_y \approx 0.25\ G$$
$$B_z \approx 0.38\ G$$

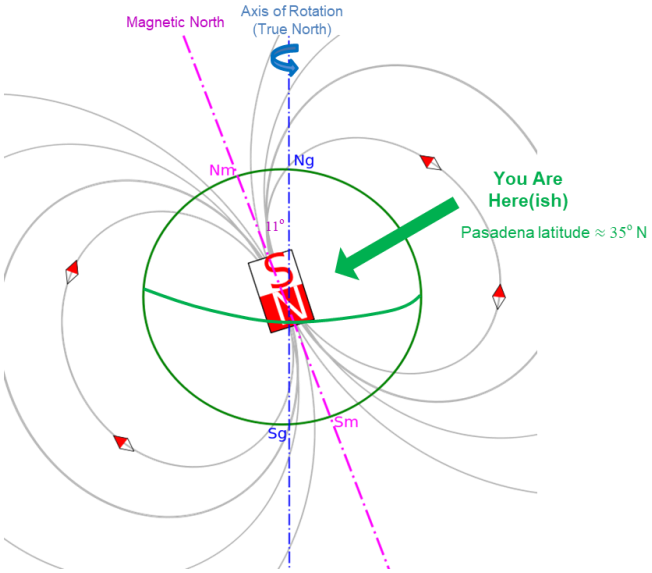


Figure 5: The earth's magnetic field can be approximated as that from a giant dipole.

As the above figure also shows, the Earth Field approximation was consistent with the magnetic field that I found in my office, as measured with a Hall probe.

The field in the lab was a different story. Initially, I used my phone's built-in magnetometer to estimate the environmental field in the lab and found results that varied dramatically with location. Further examination with a Hall probe revealed that the field strength grew as we lowered the probe to the floor, reaching magnitudes as large as 13 Gauss.

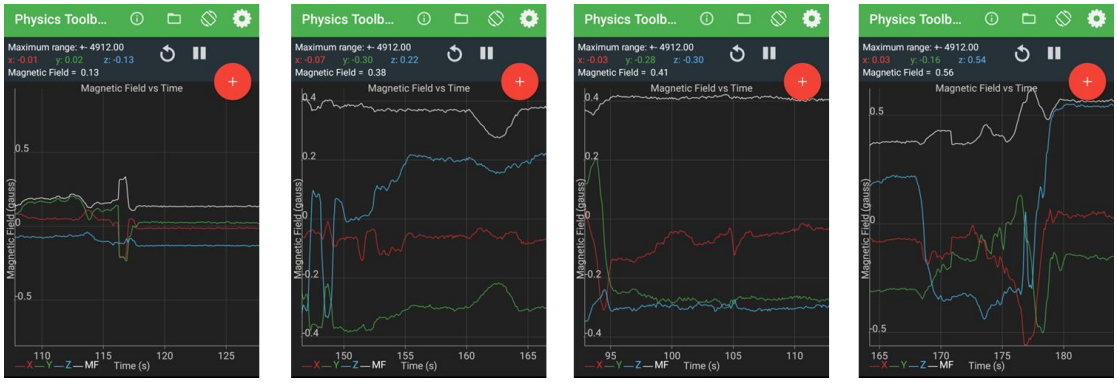


Figure 6: Timestreams of magnetic field measurements at different locations in the lab, as measured by my phone's built in magnetometer. The abrupt time-dependent changes are due to to movements/rotations of my phone; the important measurements are at the end of each timestream, when my phone was held steadily.

2.2. Preliminary Scans

Since the field was strongest at floor level, we hypothesized that the non-ideal earth field was likely caused by magnetization of the (iron) rebar embedded deeply into our lab's concrete floor. I tested this idea by measuring the z-component of the magnetic field at floor-level, sampling every square inch for approximately 6 square feet: Here's an intensity map of that data (right), as well as a picture of the measurements that I recorded (left):

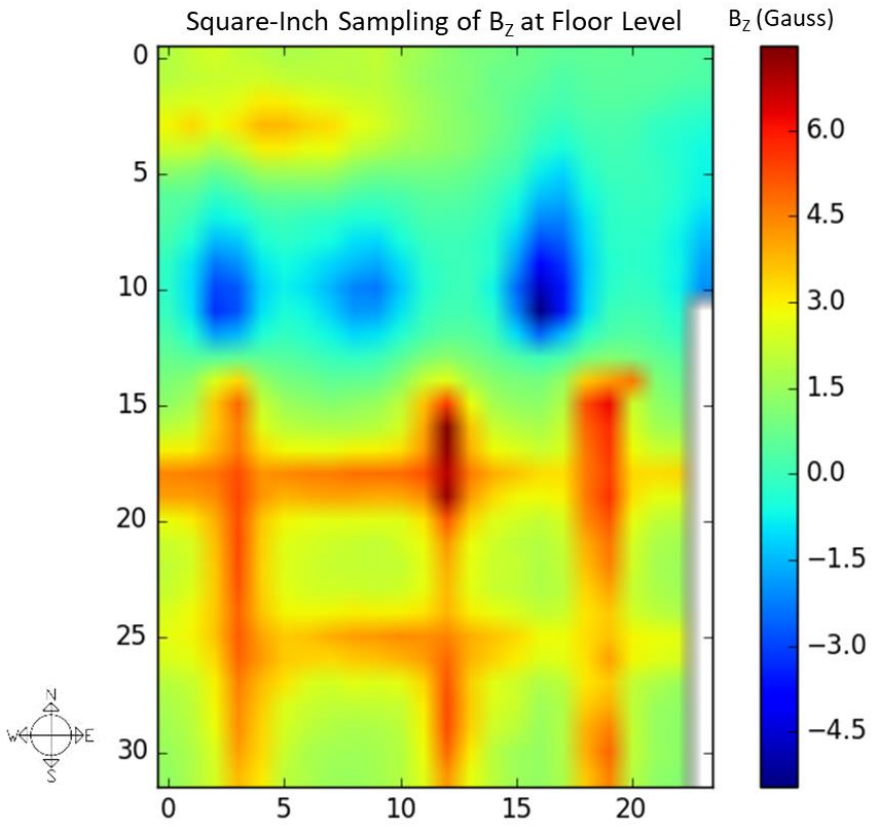


Figure 7: Floor field intensity map, where the x and y axes are in inches. The grid of highly magnetized rebar can be clearly seen in the bottom half of this map.

The rebar's grid pattern is quite apparent in the lower 2/3 of this map (though I'm still unsure what to make of the top portion). Despite the fact that I was able to use a hall probe to detect the locations of iron bars ~ a foot below the concrete, knowledge of the magnetization structure didn't prove that useful, because the rebar "grid" isn't very regularly spaced, so it's not possible to predict the locations of subsequent bars based on knowledge of observed bar locations.

The original hope was to degauss the rebar, but that was not a viable strategy due to the large extent of the magnetized rebar relative to the size of our degaussing magnet. The next plan was to simply find a better spot in the lab to set up our experiment. In order to determine "cool" spots, I used the Hall probe to scan the entire lab floor, sampling once every square foot. Here's a picture of me marking off points to be measured:



Figure 8: Me marking points to measure, spaced 1 foot apart.

The full floor intensity maps (below) clearly show the large-scale structure of the floor magnetization:

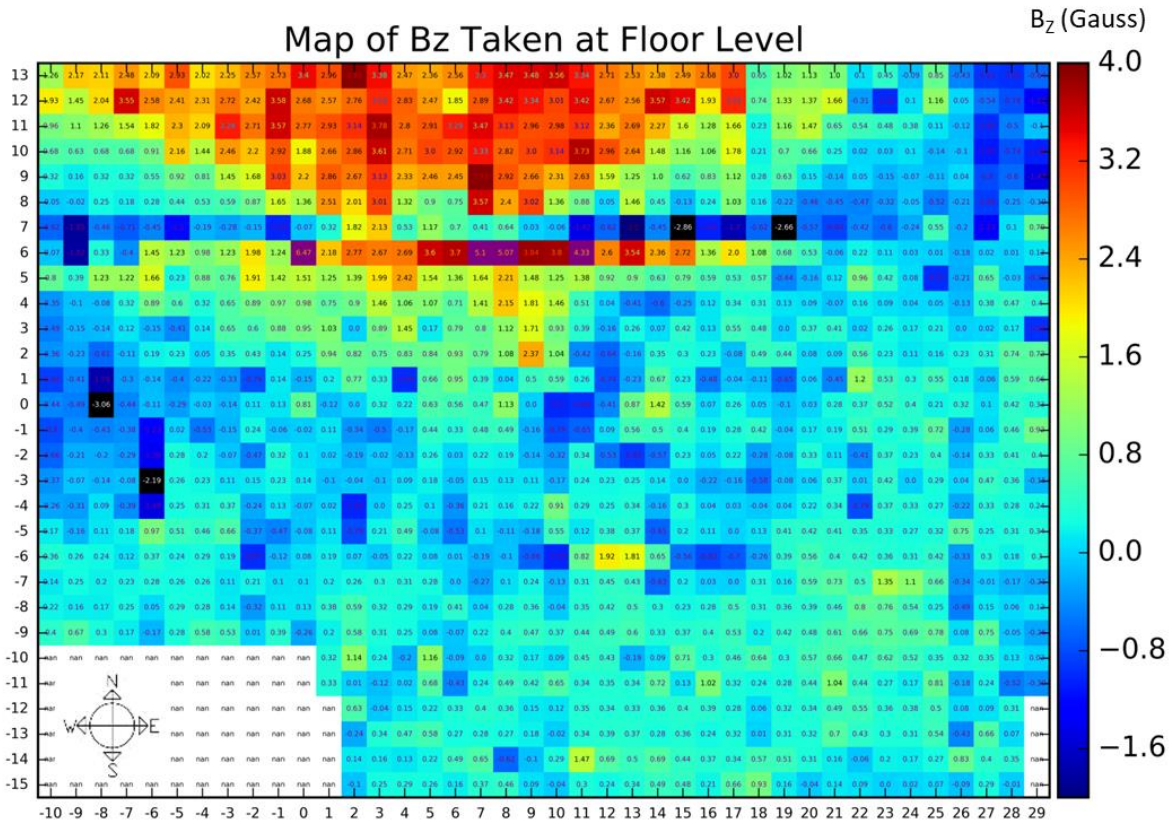


Figure 9: Floor field intensity map, where the x and y axes are in feet.

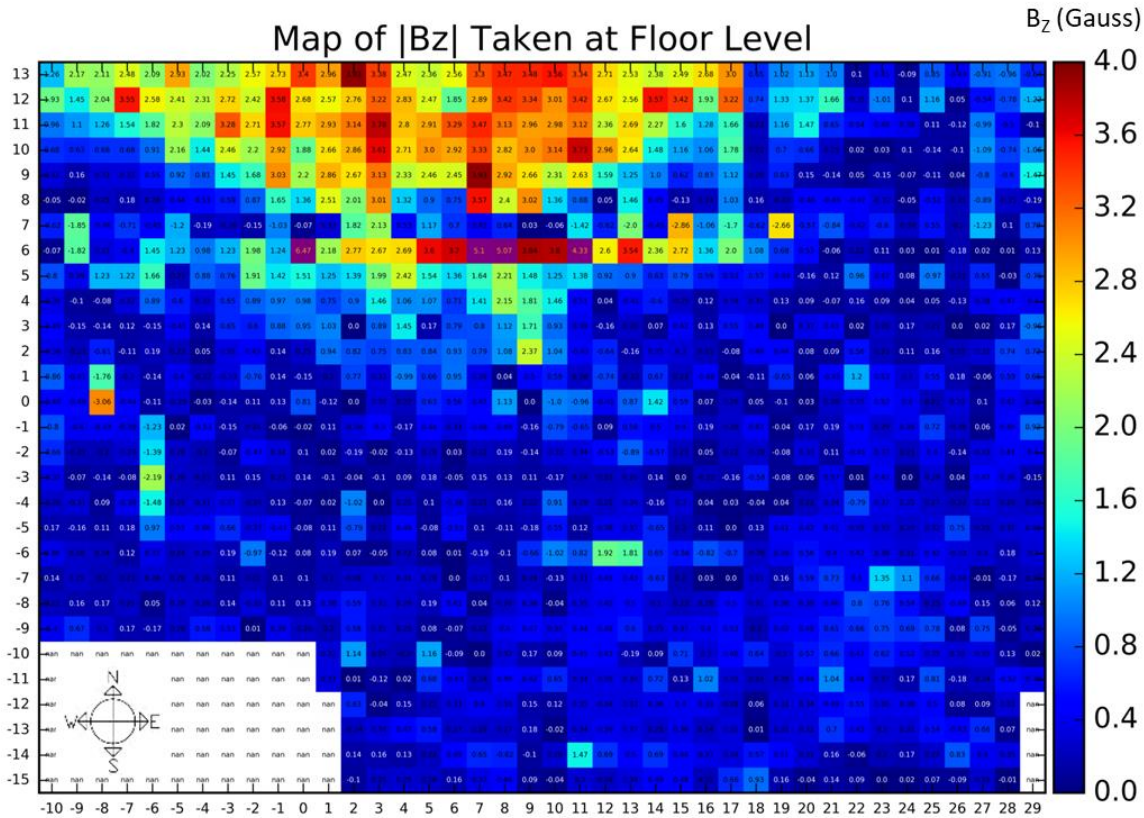


Figure 10: Absolute value plot of the data shown in figure 9.

We suspected that the floor had probably been magnetized by the Caltech Electron Synchrotron (which used to be located in the same lab), and no one had bothered to close the floor's hysteresis loop prior to decommissioning the accelerator. Recently, our group found some old photographs which supported our suspicion:



Figure 11 [11]: Old photographs of the synchrotron

I tried overlaying the above left photograph with my intensity map, to moderate success. I wasn't able to recreate the perspective perfectly, but the below figure should show how the field from the southernmost part of the synchrotron's storage ring resulted in the large rebar magnetization that we observe in the northernmost part our lab's floor space.

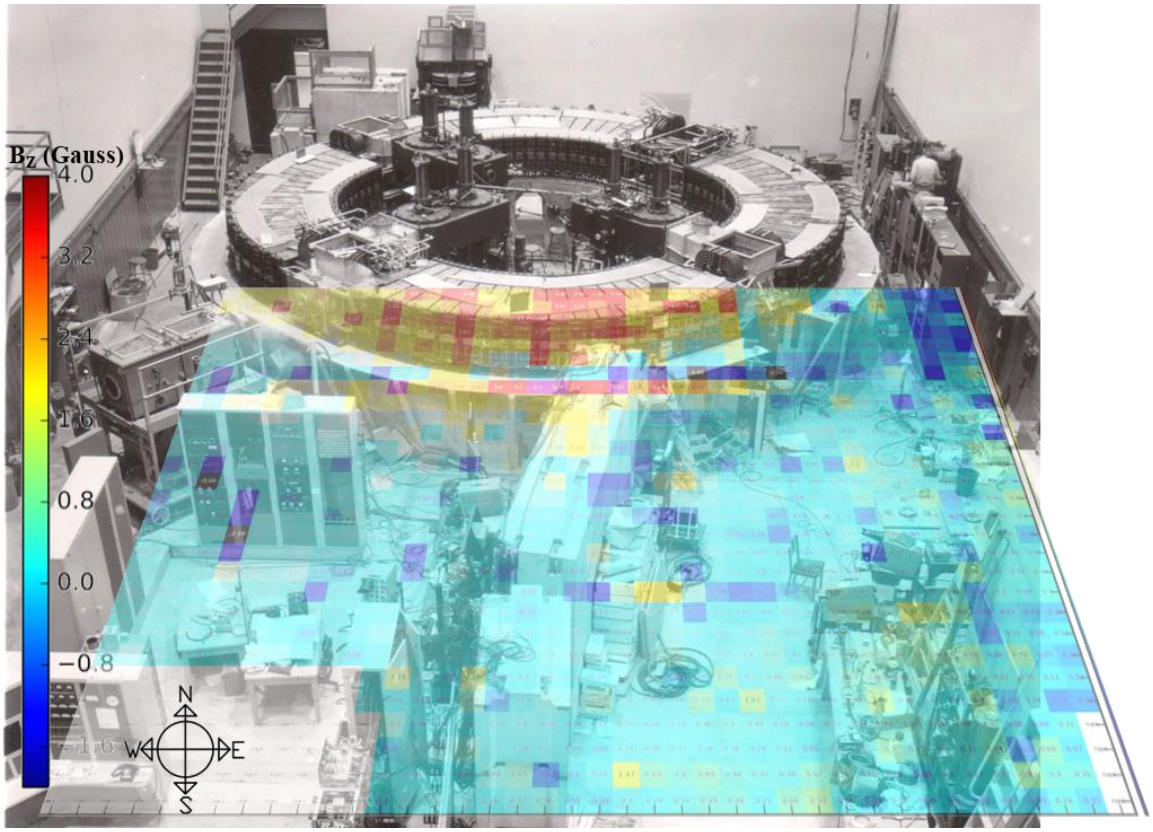


Figure 12: Projection of B_z floor map onto one of the old synchrotron photographs. I was unable to line up the perspectives perfectly, but the overlap is still quite compelling.

2.3. Relocation

From the full floor B_z map, we see that the environmental field is least ideal in the northernmost part of our available floor space, but that the floor magnetization drops off substantially as you move south. Although there is additional floor space further south than what I have mapped, the space is currently occupied by the third-scale model of the nEDM apparatus. Using this, we had a few possible relocation strategies:

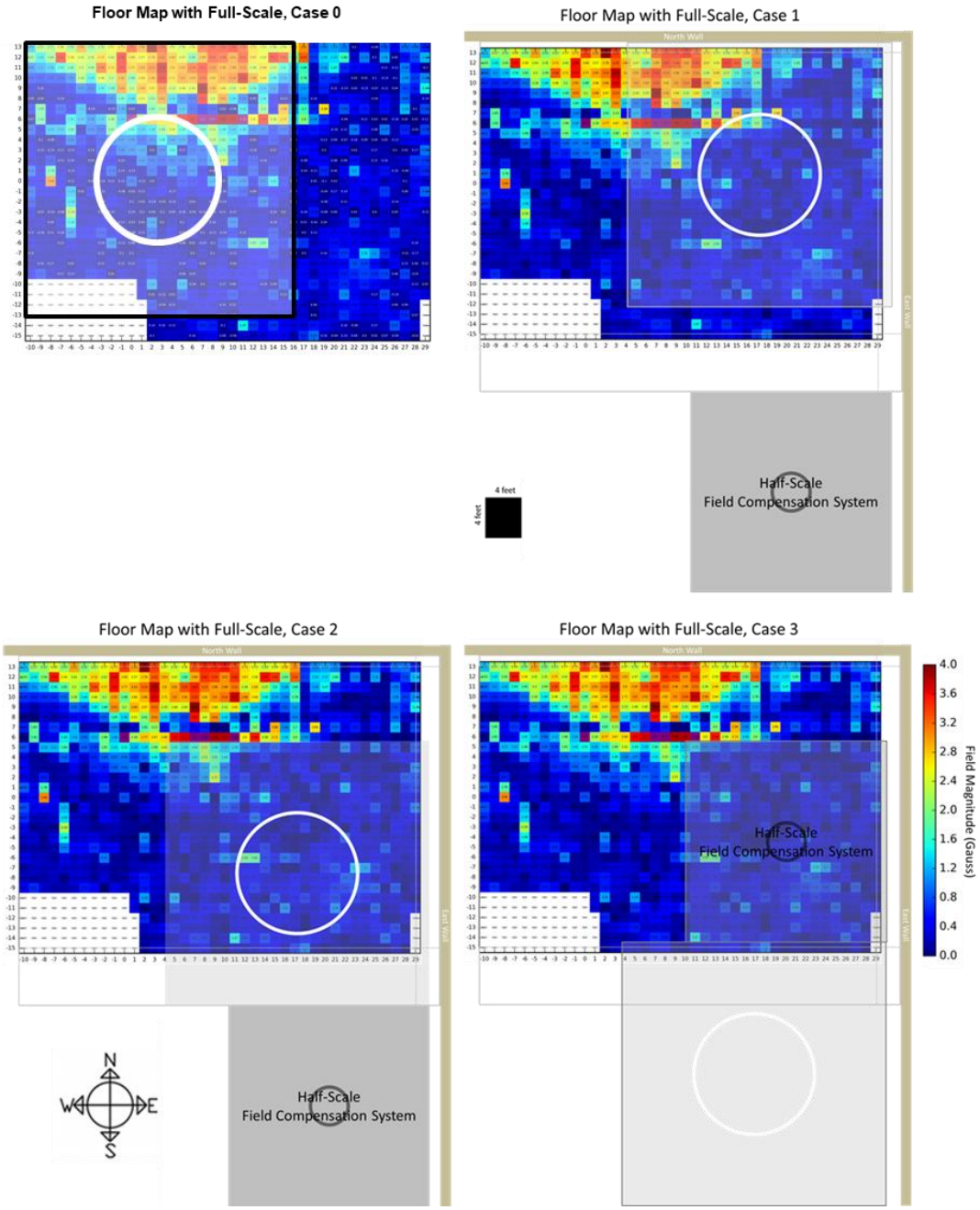


Figure 13: Possible floor layouts for the nEDM experiment. The big square and circle represent the full scale FCS and Mu-Metal shielding. The smaller square and circle correspond to the third-scale apparatus.

Case 0 reflects the originally intended location of the nEDM apparatus, and it is clearly sub-optimal due to the magnetic shielding's proximity to the most-heavily magnetized section of floor. Case 1 would have offered a slight improvement, but we could still get further away from the largest hot spot by moving south. While the Case 3 positioning would have mitigated the effect of floor magnetization the most, it would require moving the (already constructed and operational) third-scale apparatus – likely a month-long endeavor. Instead, we decided to further investigate the viability of Case 2.

2.4. Fluxgate Magnetometer 3D Map

From the large scale floor map, we knew that – in the area that would be enclosed by the outermost layer of magnetic shielding as per Case 2 – the vertical field at floor level is very similar to the Earth field. There was concern about how the field shape might change with height, however, and if the rebar-induced field might have a significant non-vertical component at locations displaced from the region of maximal magnetization. To investigate these concerns, I used a 3-axis fluxgate magnetometer to sample the entire 3-component field at many locations within the region of greatest interest. Specifically, I circumscribed the 12 ft diameter Mu-Metal cross-section with a 12 ft square, and then selected 16 equally spaced (x, y) coordinates within the square's area. At each selected coordinate, I used the fluxgate to measure the B_x, B_y, and B_z components of the environmental magnetic field at 14 equally-spaced heights, ranging from 0.5 ft up to 7 ft above ground level.

At each (x, y) coordinate, I recorded a time stream of the 3-component data, sampled once per second. While recording data, I incremented the fluxgate height by 0.5 ft approximately every 30 seconds.:



Figure 14: Three-axis fluxgate magnetometer, attached to a pole with every half-foot marked off.

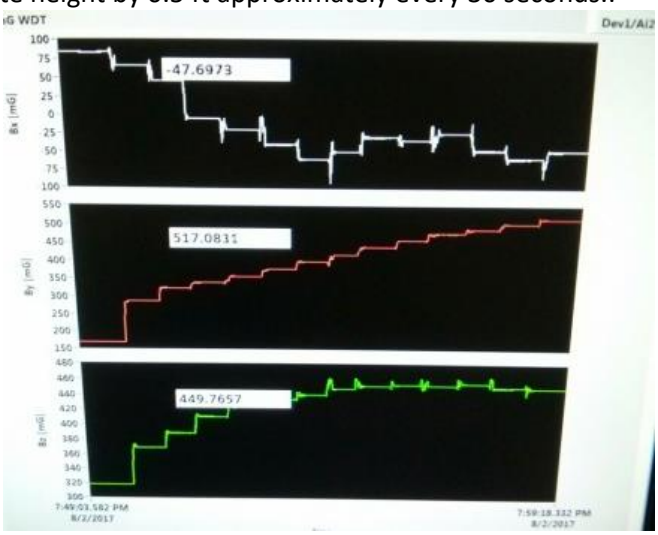


Figure 15: Example timestream of fluxgate data.

These time streams were manually parsed in order to isolate the data points for each specific height, allowing us to extract mean values and standard deviations for each magnetic field component at all 14 heights for every one of the 16 (x, y) coordinates sampled. The figure below shows how the environmental field varies as a function of height from the floor:

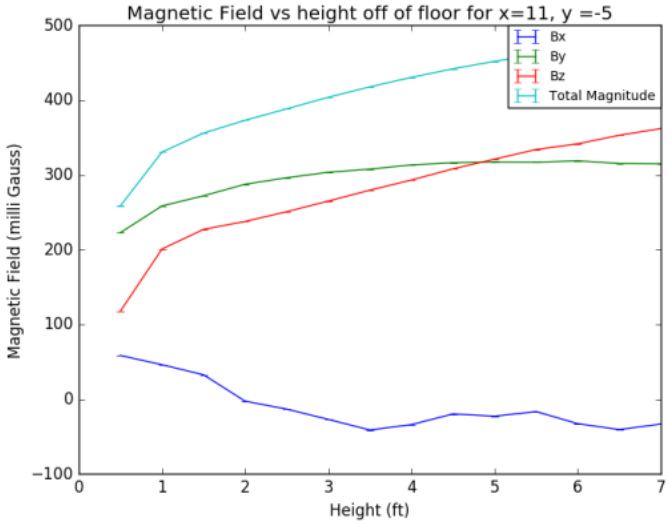


Figure 16: Magnetic field components as a function of height off floor. Interestingly, the magnitude grew with height at this point, but fortunately this was not the case for points further to the East (larger x-values).

After munging the time stream data for all 16 sampled (x, y) coordinates, I was able to plot all of the measurements as a 3D vector field, which gave me a good sense of the field's behavior in this region:

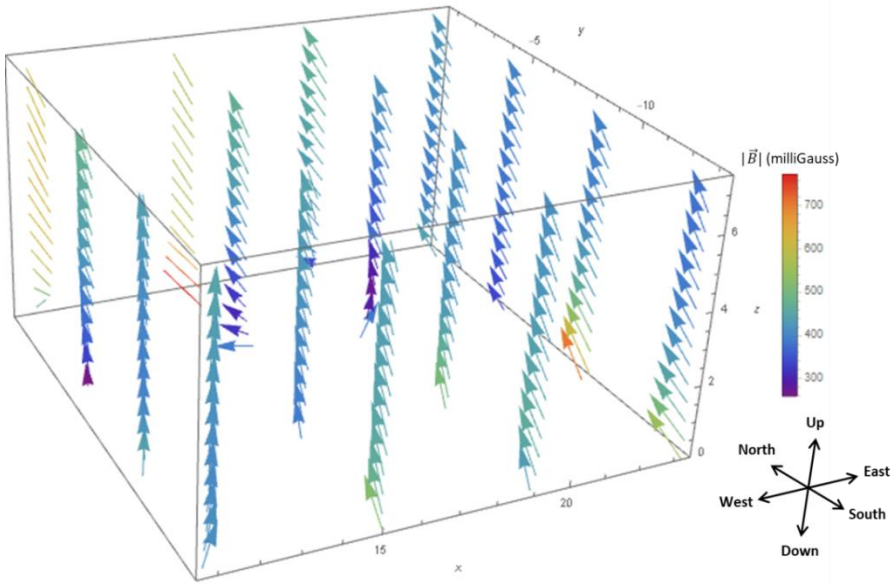


Figure 17: Vector field showing the environmental field data measured by the fluxgate. The (12ft x 12ft) base of this prism circumscribes the location of the Mu-Metal cylinder, as given by Floor Map Case 2.

As can be seen clearly in the figure above, the field is non-ideal in the northwest corner, as well as at various points close to the floor. Outside of these regions however, the field is very nearly uniform. I calculated the mean field component values from all 4x4x14 data points, and found that $< B_x > \approx 0.064\ G$, $< B_y > \approx 0.267\ G$, and $< B_z > \approx 0.312\ G$. These averages were comparable to the Earth Field values that I had previously recorded in my office, so we can conclude that the earth field is the largest environmental field source in the region specified by the Case 2 relocation plan. The effects of rebar magnetization will likely still present a slight challenge for our group however, since the floor field introduces non-negligible gradients that will need to be addressed. The vector field below is similar to the one above, except now the mean field components have been subtracted off of each vector, highlighting the non-uniformity of the environment's field:

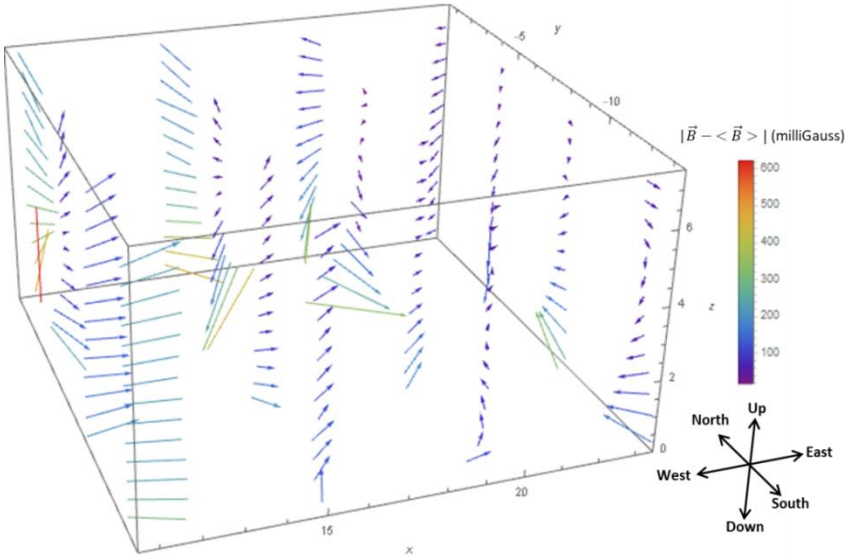


Figure 18: Vector field showing deviations from the mean in the environmental field. This shows that the field has significant nonuniformity in some locations, and these spatial gradients will need to be accounted for.

3. Field Compensation System

The Field Compensation system (FCS) is an 8m x 8m x 12ft rectangular prism, with rectangular current loops wrapped along the edges of each prism face. The largest sets of wires in the FCS are the "Direct Current Constant Offset Coils", and their function is to generate magnetic fields (Biot-Savart law) which maximally offset any permanent (time-independent) environmental fields. In order to determine the proper design specifications prior to building the apparatus (and whether or not those specifications are even feasible for the given field that needs to be offset), I simulated the FCS in Mathematica.

To begin, I used the equation for the magnetic field from a finite line of current, and wrote a function to return the Magnetic field at any (x, y, z) coordinate based on specified endpoints for a line segment of current. Next, I defined a function to superpose the fields from 4 line segments of equal current, so as to return the field from a rectangular current loop, with the rectangle's vertices given as inputs.

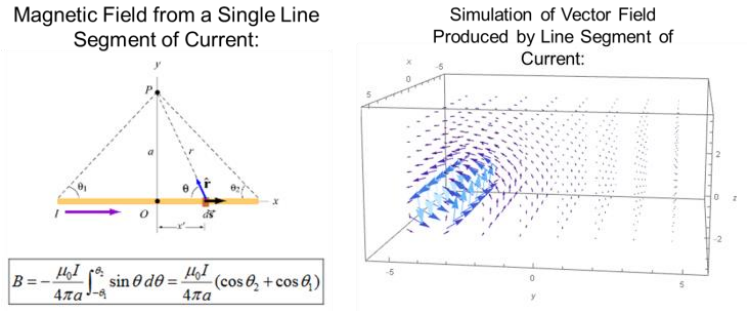


Figure 19: Equation for magnetic field at point P from a line segment of current and simulation of the corresponding magnetic field .

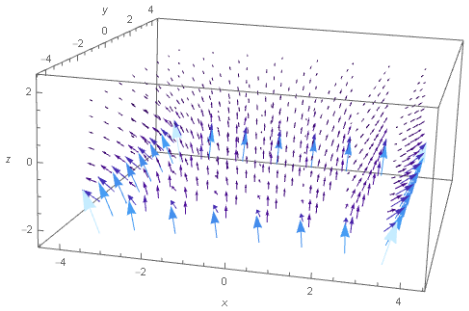


Figure 20: Magnetic field from rectangular current loop

From this, I defined a third function that superposed the fields from 6 rectangular current loops (one for each face of the prison), each oriented along one face of the prism, and with its own adjustable current.

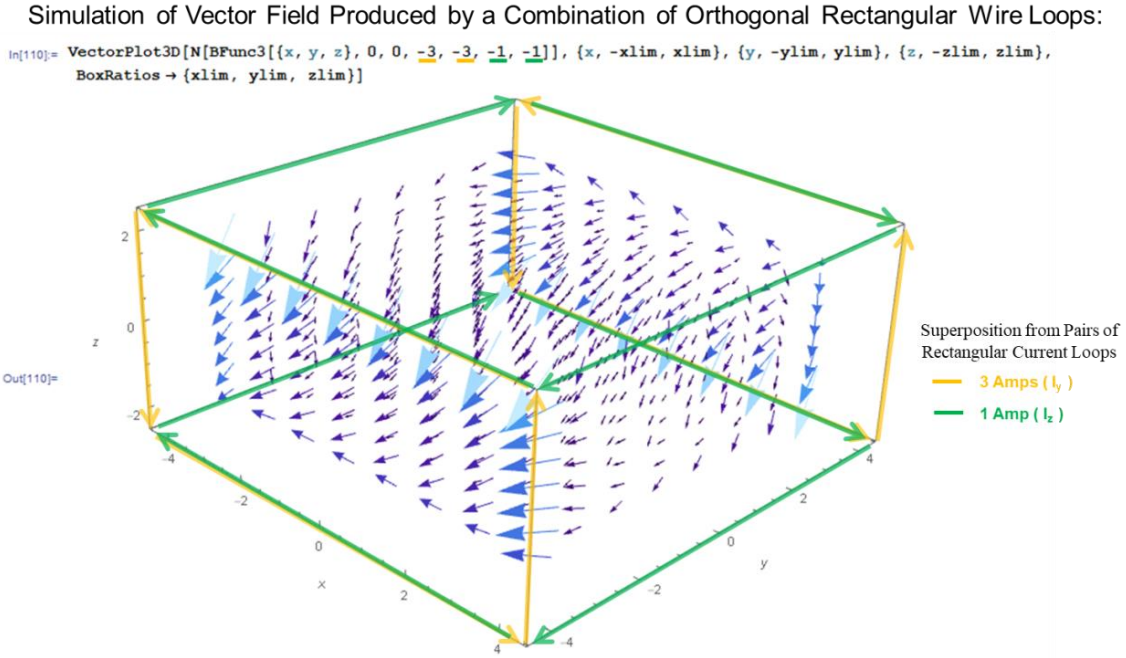


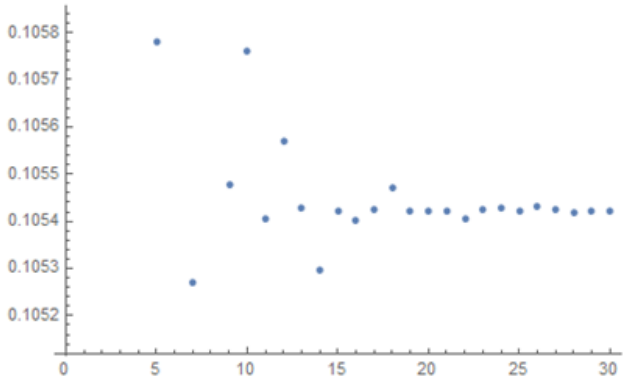
Figure 21: Magnetic field generated from a superposition of rectangular current loops

Since I wanted to approximate the configuration of currents that would be required to minimize the magnetic flux passing through our magnetic shielding, I wrote functions to superpose my six-rectangle-loop field with a uniform environmental field, then estimate the mean magnitude of this net-field along the cylindrical surface of the outermost shielding layer. I was unable to compute the true mean field magnitude, because the integral proved too intractable for Mathematica, so instead I simply averaged the field at 3*n points along the shielding cylinder's surface, selected at the top, middle, and bottom of

the cylinder and spaced $2\,\pi/n$ radians apart. To ensure the validity of my estimate, I plotted the mean magnitude estimates for a range of n values, and then selected a sufficiently large n such that the estimates had stabilized.

```
In[177]:= magData = Table[{n, BMagAv[Ix, Ix, Iy, Iy, Iz, Iz, n]}, {n, 1, 30}];
          ListPlot[magData]
```

Out[178]=




```
In[179]:= BMagAv[Ix, Ix, Iy, Iy, Iz, Iz, 25]
Out[179]= 0.105423
```

Figure 22: Average net B-field Magnitude as a function of the number of sample points (n). As n increases, the average magnitude converges to a more stable estimate.

For the field that I wanted the FCS to optimally offset, I used the average environment field described at the end of Section 2, $< B_x > \approx 0.064\ G$, $< B_y > \approx 0.267\ G$, and $< B_z > \approx 0.312\ G$. To be safe, I increased the magnitude of each component by ~20%, so the environment field used in my simulation was $\vec{B}_{env}$ = {0.08G East, 0.32 G North, 0.373 G Up}, which looks like:

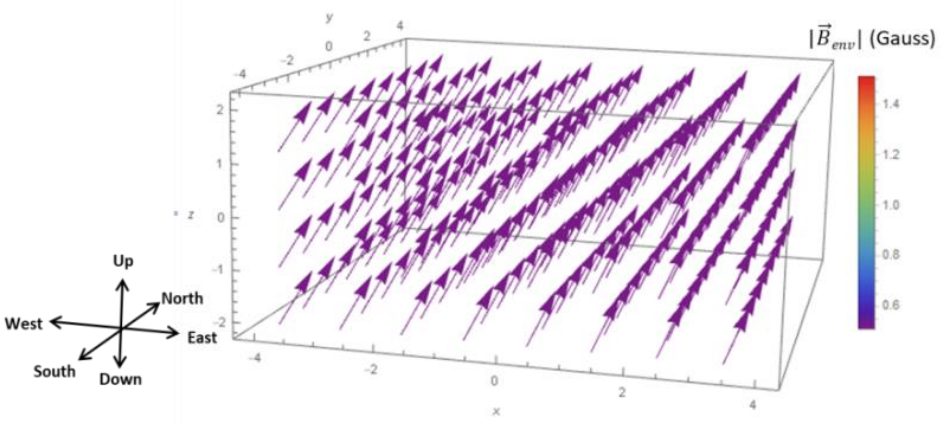


Figure 23: The ideal uniform environemental magnetic field, generated by the earth.

Next, since this is a uniform field, I assumed from the symmetry of the system (as well as the limitations of the set-up) that the currents running along opposing faces of the FCS should be equal, leaving me with only 3 free parameters to optimize over. Using Mathematica's built-in FindMinimum[] function, I was able to determine the current values (I_x, I_y, & I_z) that minimized the output of my mean-field-magnitude-estimating-function (BMagAv[]), with an n-value of 25. These optimal current selections were $\{I_x, I_y, I_z\} \approx \{83, 343, 166\}$ Amp-turns, and they result in the net-field below:

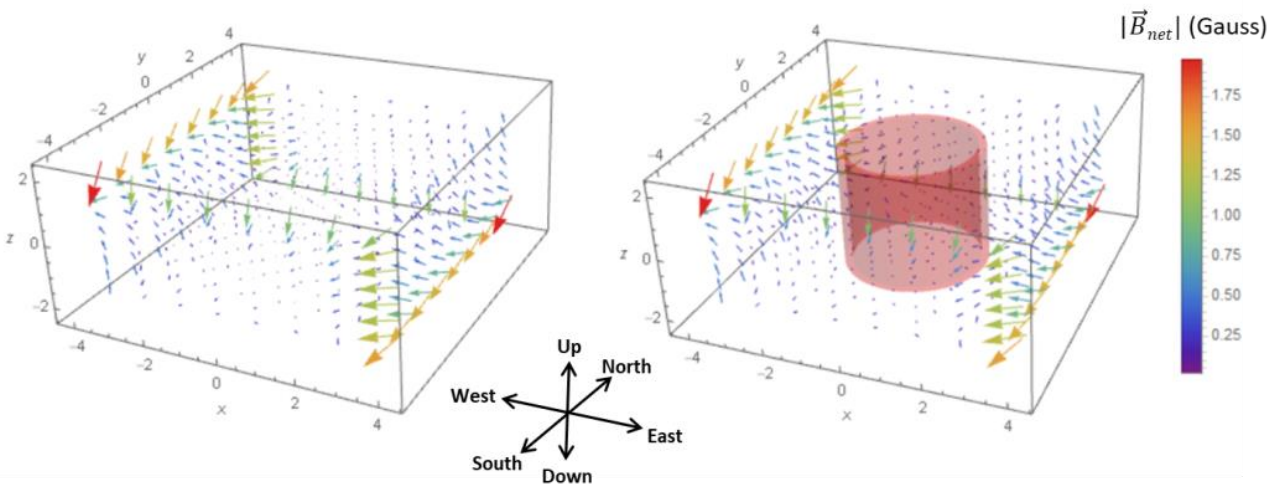


Figure 24: Simulated net (Earth + FCS) magnetic field inside the FCS. The red cylinder (right) represents the location of the Mu-Metal shielding.

The plot below shows what this field looks like only along the surface of the shielding cylinder:

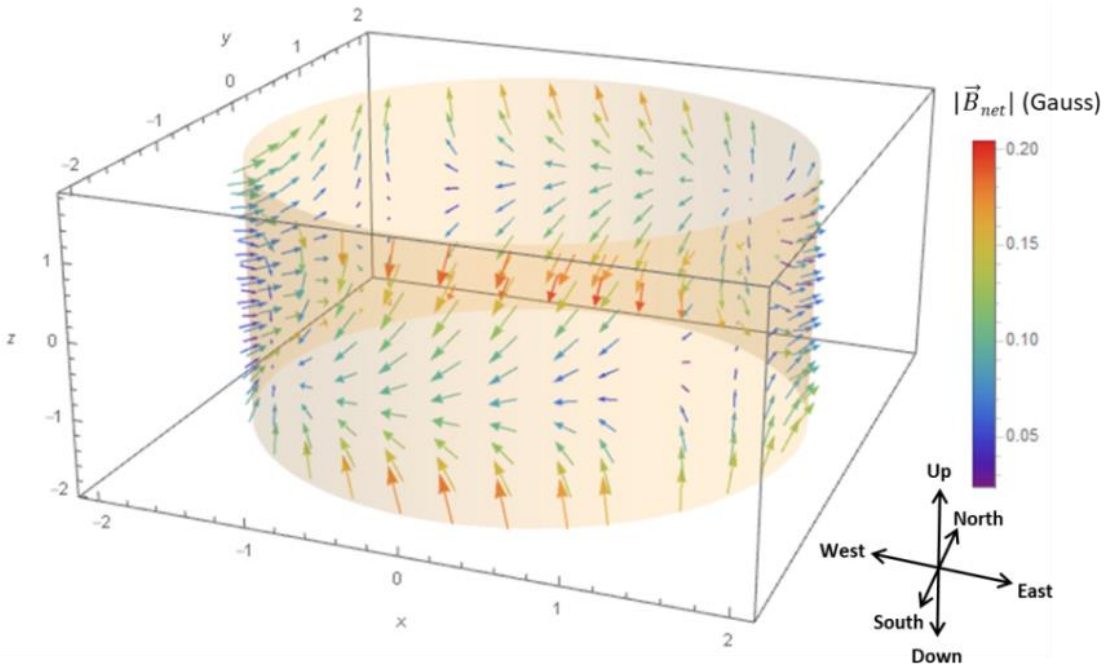


Figure 25: The net magnetic field, shown only along the surface of the shielding cylinder.

From this simulation, we see that the FCS can reduce the environmental field very effectively at the location of the Mu-metal shielding cylinder, but that it will require applying substantial currents, including a whopping 343 Amp-turns for the y-coils.

4. Coil Selection

To achieve the optimal current configurations determined in section 3, there were a number of constraints which needed to be met. First, the power supplies had maximum voltage and current levels that they could apply. Two of the power supplies were capable of applying up to 60 Volts, and as many as 15 Amps. The remaining power supply had maximum voltage and current levels of 40 Volts and 50 Amps, respectively. The DC constant offset component of the FCS is powered by just 3 power supplies, meaning the loops on opposing faces of the rectangular prism are powered by the same supply.

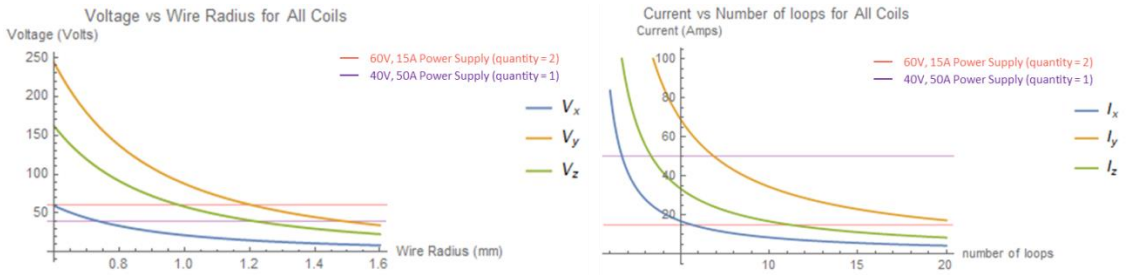


Figure 26: (Left) Voltage required to provide coils with necessary Amp-turns, as a function of wire radius. (Right) Current required to provide coils with necessary Amp-turns, as a function of the number of loops.

In addition to limitations on the maximum currents and voltages, we also wanted to impose an upper bound on the power dissipation through the loops. In the third-scale model, there was an accident where the FCS coils snapped. It's suspected that the coils were under tension, and that, when the wires became hot from joule heating, they were gradually 'exercised' via mechanical creep until they eventually snapped. Such an incident would potentially be very dangerous for the full-scale apparatus, so we want to avoid similar mishaps. To circumvent this issue, we intend to wind the coils less tightly and to limit the amount of power that the coils dissipate. To determine appropriate maximum power limits, I recorded the maximum power dissipation in the (now safe) third-scale FCS and then divided it by the length of a single loop to calculate a power-dissipation-per-length density. I then multiplied this density by the lengths of each loop component in the full-scale apparatus to estimate maximum powers that we wanted to apply through each component.

With these conditions established, I compiled a list of plausible wire gauges with corresponding prices per unit length obtained from McMaster-Carr, and wrote a Mathematica function to compute the necessary numbers of loops and associated costs for all possible permutations for the power supplies and the x, y, and z coil gauges, conditioned on the aforementioned constraints on maximum current and power dissipation. This yielded a long 2d array containing cost, current, voltage, and power output data for each x, y, z wire gauge + power supply permutation. I imported this 2d array into Python, isolated only the viable permutations (such that all of the currents and voltages met the constraints provided by their coil's respective power supplies), and then output the resulting data as sortable spreadsheets, which I will include as an appendix.

The table allowed my group to pick a set of coils based on practicality factors such as cost and the total number of loops (more loops means more man hours assembling massive coils), since the constraints imposed on the system had already been met for all of the configurations shown in the table.

5. Conclusions

Residual floor magnetization presented a serious challenge to the efficacy of the nEDM apparatus' external field mitigation system. After extensive measurement and analysis however, we were able to find a way to (largely) avoid the difficulties presented by the magnetized floor. Additionally, I created a simulation of the Field Compensation System, which (while also taking into account residual floor magnetization effects) allowed me to estimate the order of magnitude required to achieve optimal current configurations with the FCS. This provided a valuable constraint that helped us determine the wires that we needed to purchase in order to construct the FCS.

6. Acknowledgements

I would like to thank Professor Filippone for giving me the opportunity to research with the nEDM group and providing me with the guidance that I needed to carry out my research. I would also like to thank Simon Slutsky for providing constructive feedback and suggestions on my work. I'm grateful to Charles (Chub) Osthelder for his incredibly swift construction of the height-adjustable stand used for the3D fluxgate measurements, as well as a good number of entertaining conversations. Lastly, thank you Bob Carr, Wanchun Wei, Chris Swank, and Simon (again) for sharing many enjoyable lunch breaks with me.

7. Appendix: Wire Configuration Tables

The tables below shows viable wire configurations subject to limitations on the maximum linear power density that can be applied through each coil. Every row represents a potential FCS configuration, and contains (1) the material cost of the configuration in USD, as well as (for each component x, y, and z) the: (2-4) wire gauges, (5-7) voltages, (8-10) currents, (11-13) the number of loops, (14-16) outer diameters of single loops, (17-19) cross sectional areas of whole bundles, (20-22) power dissipations, and (23-25) total lengths of wire.

Table 1 Limit set to maximum power density currently being applied to the third-scale model coils (7.3W/m):

#totalCost	#GaugeX	#GaugeY	#GaugeZ	#voltx(V)	#volty(V)	#voltz(V)	#currx(A)	#curry(A)	#currz(A)	#nx	#ny	#nz	#xOD(in)	#yOD(in)	#zOD(in)	#xCSA(in^	#yCSA(in^	#zCSA(in^	#xPow(W)	#yPow(W)	#zPow(W)	#wlenX(ft	#wlenY(ft	#wlenZ(ft
2446.09	14	10	12	33.45	51.94	52.96	9.3	-6.48	-8.76	9	53	19	0.11	0.16	0.13	0.17	2.13	0.5	311.17	336.49	463.76	1376.88	8108.31	3989.51
2473.81	14	10	10	33.45	51.94	34.55	9.3	-6.48	-12.8	9	53	13	0.11	0.16	0.16	0.17	2.13	0.52	311.17	336.49	442.18	1376.88	8108.31	2729.66
2648.74	14	10	8	33.45	51.94	21.45	9.3	-6.48	-13.87	9	53	12	0.11	0.16	0.21	0.17	2.13	0.83	311.17	336.49	297.36	1376.88	8108.31	2519.69
2314.89	14	8	12	33.45	32.24	52.96	9.3	-10.4	-8.76	9	33	19	0.11	0.21	0.13	0.17	2.29	0.5	311.17	335.47	463.76	1376.88	5048.57	3989.51
2342.61	14	8	10	33.45	32.24	34.55	9.3	-10.4	-12.8	9	33	13	0.11	0.21	0.16	0.17	2.29	0.52	311.17	335.47	442.18	1376.88	5048.57	2729.66
2517.55	14	8	8	33.45	32.24	21.45	9.3	-10.4	-13.87	9	33	12	0.11	0.21	0.21	0.17	2.29	0.83	311.17	335.47	297.36	1376.88	5048.57	2519.69
2428.94	12	10	12	19.42	51.94	52.96	16.75	-6.48	-8.76	5	53	19	0.13	0.16	0.13	0.13	2.13	0.5	325.15	336.49	463.76	764.93	8108.31	3989.51
2456.66	12	10	10	19.42	51.94	34.55	16.75	-6.48	-12.8	5	53	13	0.13	0.16	0.16	0.13	2.13	0.52	325.15	336.49	442.18	764.93	8108.31	2729.66
2631.59	12	10	8	19.42	51.94	21.45	16.75	-6.48	-13.87	5	53	12	0.13	0.16	0.21	0.13	2.13	0.83	325.15	336.49	297.36	764.93	8108.31	2519.69
2297.75	12	8	12	19.42	32.24	52.96	16.75	-10.4	-8.76	5	33	19	0.13	0.21	0.13	0.13	2.29	0.5	325.15	335.47	463.76	764.93	5048.57	3989.51
2325.47	12	8	10	19.42	32.24	34.55	16.75	-10.4	-12.8	5	33	13	0.13	0.21	0.16	0.13	2.29	0.52	325.15	335.47	442.18	764.93	5048.57	2729.66
2500.4	12	8	8	19.42	32.24	21.45	16.75	-10.4	-13.87	5	33	12	0.13	0.21	0.21	0.13	2.29	0.83	325.15	335.47	297.36	764.93	5048.57	2519.69
2456.79	10	10	12	12.67	51.94	52.96	20.93	-6.48	-8.76	4	53	19	0.16	0.16	0.13	0.16	2.13	0.5	265.14	336.49	463.76	611.95	8108.31	3989.51
2484.51	10	10	10	12.67	51.94	34.55	20.93	-6.48	-12.8	4	53	13	0.16	0.16	0.16	0.16	2.13	0.52	265.14	336.49	442.18	611.95	8108.31	2729.66
2659.44	10	10	8	12.67	51.94	21.45	20.93	-6.48	-13.87	4	53	12	0.16	0.16	0.21	0.16	2.13	0.83	265.14	336.49	297.36	611.95	8108.31	2519.69
2325.59	10	8	12	12.67	32.24	52.96	20.93	-10.4	-8.76	4	33	19	0.16	0.21	0.13	0.16	2.29	0.5	265.14	335.47	463.76	611.95	5048.57	3989.51
2353.31	10	8	10	12.67	32.24	34.55	20.93	-10.4	-12.8	4	33	13	0.16	0.21	0.16	0.16	2.29	0.52	265.14	335.47	442.18	611.95	5048.57	2729.66
2528.25	10	8	8	12.67	32.24	21.45	20.93	-10.4	-13.87	4	33	12	0.16	0.21	0.21	0.16	2.29	0.83	265.14	335.47	297.36	611.95	5048.57	2519.69
2420.87	8	10	12	7.86	51.94	52.96	41.87	-6.48	-8.76	2	53	19	0.21	0.16	0.13	0.14	2.13	0.5	329.18	336.49	463.76	305.97	8108.31	3989.51
2448.59	8	10	10	7.86	51.94	34.55	41.87	-6.48	-12.8	2	53	13	0.21	0.16	0.16	0.14	2.13	0.52	329.18	336.49	442.18	305.97	8108.31	2729.66
2623.52	8	10	8	7.86	51.94	21.45	41.87	-6.48	-13.87	2	53	12	0.21	0.16	0.21	0.14	2.13	0.83	329.18	336.49	297.36	305.97	8108.31	2519.69
2289.68	8	8	12	7.86	32.24	52.96	41.87	-10.4	-8.76	2	33	19	0.21	0.21	0.13	0.14	2.29	0.5	329.18	335.47	463.76	305.97	5048.57	3989.51
2317.4	8	8	10	7.86	32.24	34.55	41.87	-10.4	-12.8	2	33	13	0.21	0.21	0.16	0.14	2.29	0.52	329.18	335.47	442.18	305.97	5048.57	2729.66
2492.33	8	8	8	7.86	32.24	21.45	41.87	-10.4	-13.87	2	33	12	0.21	0.21	0.21	0.14	2.29	0.83	329.18	335.47	297.36	305.97	5048.57	2519.69
2446.09	14	10	12	33.45	51.94	52.96	9.3	-6.48	-8.76	9	53	19	0.11	0.16	0.13	0.17	2.13	0.5	311.17	336.49	463.76	1376.88	8108.31	3989.51
2473.81	14	10	10	33.45	51.94	34.55	9.3	-6.48	-12.8	9	53	13	0.11	0.16	0.16	0.17	2.13	0.52	311.17	336.49	442.18	1376.88	8108.31	2729.66
2648.74	14	10	8	33.45	51.94	21.45	9.3	-6.48	-13.87	9	53	12	0.11	0.16	0.21	0.17	2.13	0.83	311.17	336.49	297.36	1376.88	8108.31	2519.69
2314.89	14	8	12	33.45	32.24	52.96	9.3	-10.4	-8.76	9	33	19	0.11	0.21	0.13	0.17	2.29	0.5	311.17	335.47	463.76	1376.88	5048.57	3989.51
2342.61	14	8	10	33.45	32.24	34.55	9.3	-10.4	-12.8	9	33	13	0.11	0.21	0.16	0.17	2.29	0.52	311.17	335.47	442.18	1376.88	5048.57	2729.66
2517.55	14	8	8	33.45	32.24	21.45	9.3	-10.4	-13.87	9	33	12	0.11	0.21	0.21	0.17	2.29	0.83	311.17	335.47	297.36	1376.88	5048.57	2519.69
2451.77	12	10	12	19.42	51.94	52.96	13.96	-6.48	-8.76	6	53	19	0.13	0.16	0.13	0.16	2.13	0.5	270.95	336.49	463.76	917.92	8108.31	3989.51
2479.49	12	10	10	19.42	51.94	34.55	13.96	-6.48	-12.8	6	53	13	0.13	0.16	0.16	0.16	2.13	0.52	270.95	336.49	442.18	917.92	8108.31	2729.66
2654.42	12	10	8	19.42	51.94	21.45	13.96	-6.48	-13.87	6	53	12	0.13	0.16	0.21	0.16	2.13	0.83	270.95	336.49	297.36	917.92	8108.31	2519.69
2320.58	12	8	12	19.42	32.24	52.96	13.96	-10.4	-8.76	6	33	19	0.13	0.21	0.13	0.16	2.29	0.5	270.95	335.47	463.76	917.92	5048.57	3989.51
2348.3	12	8	10	19.42	32.24	34.55	13.96	-10.4	-12.8	6	33	13	0.13	0.21	0.16	0.16	2.29	0.52	270.95	335.47	442.18	917.92	5048.57	2729.66
2523.23	12	8	8	19.42	32.24	21.45	13.96	-10.4	-13.87	6	33	12	0.13	0.21	0.21	0.16	2.29	0.83	270.95	335.47	297.36	917.92	5048.57	2519.69
2527.77	10	10	12	12.67	51.94	52.96	13.96	-6.48	-8.76	6	53	19	0.16	0.16	0.13	0.24	2.13	0.5	176.76	336.49	463.76	917.92	8108.31	3989.51
2555.49	10	10	10	12.67	51.94	34.55	13.96	-6.48	-12.8	6	53	13	0.16	0.16	0.16	0.24	2.13	0.52	176.76	336.49	442.18	917.92	8108.31	2729.66
2730.42	10	10	8	12.67	51.94	21.45	13.96	-6.48	-13.87	6	53	12	0.16	0.16	0.21	0.24	2.13	0.83	176.76	336.49	297.36	917.92	8108.31	2519.69
2396.58	10	8	12	12.67	32.24	52.96	13.96	-10.4	-8.76	6	33	19	0.16	0.21	0.13	0.24	2.29	0.5	176.76	335.47	463.76	917.92	5048.57	3989.51
2424.3	10	8	10	12.67	32.24	34.55	13.96	-10.4	-12.8	6	33	13	0.16	0.21	0.16	0.24	2.29	0.52	176.76	335.47	442.18	917.92	5048.57	2729.66
2599.23	10	8	8	12.67	32.24	21.45	13.96	-10.4	-13.87	6	33	12	0.16	0.21	0.21	0.24	2.29	0.83	176.76	335.47	297.36	917.92	5048.57	2519.69
2632.98	8	10	12	7.86	51.94	52.96	13.96	-6.48	-8.76	6	53	19	0.21	0.16	0.13	0.42	2.13	0.5	109.73	336.49	463.76	917.92	8108.31	3989.51
2660.71	8	10	10	7.86	51.94	34.55	13.96	-6.48	-12.8	6	53	13	0.21	0.16	0.16	0.42	2.13	0.52	109.73	336.49	442.18	917.92	8108.31	2729.66
2835.64	8	10	8	7.86	51.94	21.45	13.96	-6.48	-13.87	6	53	12	0.21	0.16	0.21	0.42	2.13	0.83	109.73	336.49	297.36	917.92	8108.31	2519.69
2501.79	8	8	12	7.86	32.24	52.96	13.96	-10.4	-8.76	6	33	19	0.21	0.21	0.13	0.42	2.29	0.5	109.73	335.47	463.76	917.92	5048.57	3989.51
2529.51	8	8	10	7.86	32.24	34.55	13.96	-10.4	-12.8	6	33	13	0.21	0.21	0.16	0.42	2.29	0.52	109.73	335.47	442.18	917.92	5048.57	2729.66
2704.44	8	8	8	7.86	32.24	21.45	13.96	-10.4	-13.87	6	33	12	0.21	0.21	0.21	0.42	2.29	0.83	109.73	335.47	297.36	917.92	5048.57	2519.69
2446.09	14	10	12	33.45	51.94	52.96	9.3	-6.48	-8.76	9	53	19	0.11	0.16	0.13	0.17	2.13	0.5	311.17	336.49	463.76	1376.88	8108.31	3989.51
2473.81	14	10	10	33.45	51.94	34.55	9.3	-6.48	-12.8	9	53	13	0.11	0.16	0.16	0.17	2.13	0.52	311.17	336.49	442.18	1376.88	8108.31	2729.66

Table 1: Maximum power density = 7.3W/m

Table 2 Limit set to twice as large as the limit used in table 1. (14.6W/m):

#totalCost	#GaugeX	#GaugeY	#GaugeZ	#voltx(V)	#volty(V)	#voltz(V)	#currx(A)	#curry(A)	#currz(A)	#nx	#ny	#nz	#xOD(in)	#yOD(in)	#zOD(in)	#xCSA(in^	#yCSA(in^	#zCSA(in^	#xPow(W)	#yPow(W)	#zPow(W)	#wlenX(ft	#wlenY(ft	#wlenZ(ft
1305.15	14	10	12	33.45	51.94	52.96	16.75	-12.72	-13.87	5	27	12	0.11	0.16	0.13	0.1	1.09	0.32	560.11	660.52	734.29	764.93	4130.65	2519.69
1457.15	14	10	10	33.45	51.94	34.55	16.75	-12.72	-13.87	5	27	12	0.11	0.16	0.16	0.1	1.09	0.48	560.11	660.52	479.03	764.93	4130.65	2519.69
1667.58	14	10	8	33.45	51.94	21.45	16.75	-12.72	-13.87	5	27	12	0.11	0.16	0.21	0.1	1.09	0.83	560.11	660.52	297.36	764.93	4130.65	2519.69
1566.49	14	8	12	33.45	32.24	52.96	16.75	-14.93	-13.87	5	23	12	0.11	0.21	0.13	0.1	1.59	0.32	560.11	481.33	734.29	764.93	3518.7	2519.69
1718.5	14	8	10	33.45	32.24	34.55	16.75	-14.93	-13.87	5	23	12	0.11	0.21	0.16	0.1	1.59	0.48	560.11	481.33	479.03	764.93	3518.7	2519.69
1928.92	14	8	8	33.45	32.24	21.45	16.75	-14.93	-13.87	5	23	12	0.11	0.21	0.21	0.1	1.59	0.83	560.11	481.33	297.36	764.93	3518.7	2519.69
1300.69	12	10	12	19.42	51.94	52.96	27.91	-12.72	-13.87	3	27	12	0.13	0.16	0.13	0.08	1.09	0.32	541.91	660.52	734.29	458.96	4130.65	2519.69
1452.7	12	10	10	19.42	51.94	34.55	27.91	-12.72	-13.87	3	27	12	0.13	0.16	0.16	0.08	1.09	0.48	541.91	660.52	479.03	458.96	4130.65	2519.69
1663.13	12	10	8	19.42	51.94	21.45	27.91	-12.72	-13.87	3	27	12	0.13	0.16	0.21	0.08	1.09	0.83	541.91	660.52	297.36	458.96	4130.65	2519.69
1562.04	12	8	12	19.42	32.24	52.96	27.91	-14.93	-13.87	3	23	12	0.13	0.21	0.13	0.08	1.59	0.32	541.91	481.33	734.29	458.96	3518.7	2519.69
1714.04	12	8	10	19.42	32.24	34.55	27.91	-14.93	-13.87	3	23	12	0.13	0.21	0.16	0.08	1.59	0.48	541.91	481.33	479.03	458.96	3518.7	2519.69
1924.47	12	8	8	19.42	32.24	21.45	27.91	-14.93	-13.87	3	23	12	0.13	0.21	0.21	0.08	1.59	0.83	541.91	481.33	297.36	458.96	3518.7	2519.69
1303.2	10	10	12	12.67	51.94	52.96	41.87	-12.72	-13.87	2	27	12	0.16	0.16	0.13	0.08	1.09	0.32	530.28	660.52	734.29	305.97	4130.65	2519.69
1455.21	10	10	10	12.67	51.94	34.55	41.87	-12.72	-13.87	2	27	12	0.16	0.16	0.16	0.08	1.09	0.48	530.28	660.52	479.03	305.97	4130.65	2519.69
1665.64	10	10	8	12.67	51.94	21.45	41.87	-12.72	-13.87	2	27	12	0.16	0.16	0.21	0.08	1.09	0.83	530.28	660.52	297.36	305.97	4130.65	2519.69
1564.54	10	8	12	12.67	32.24	52.96	41.87	-14.93	-13.87	2	23	12	0.16	0.21	0.13	0.08	1.59	0.32	530.28	481.33	734.29	305.97	3518.7	2519.69
1716.55	10	8	10	12.67	32.24	34.55	41.87	-14.93	-13.87	2	23	12	0.16	0.21	0.16	0.08	1.59	0.48	530.28	481.33	479.03	305.97	3518.7	2519.69
1926.98	10	8	8	12.67	32.24	21.45	41.87	-14.93	-13.87	2	23	12	0.16	0.21	0.21	0.08	1.59	0.83	530.28	481.33	297.36	305.97	3518.7	2519.69
1338.27	8	10	12	7.86	51.94	52.96	41.87	-12.72	-13.87	2	27	12	0.21	0.16	0.13	0.14	1.09	0.32	329.18	660.52	734.29	305.97	4130.65	2519.69
1490.28	8	10	10	7.86	51.94	34.55	41.87	-12.72	-13.87	2	27	12	0.21	0.16	0.16	0.14	1.09	0.48	329.18	660.52	479.03	305.97	4130.65	2519.69
1700.71	8	10	8	7.86	51.94	21.45	41.87	-12.72	-13.87	2	27	12	0.21	0.16	0.21	0.14	1.09	0.83	329.18	660.52	297.36	305.97	4130.65	2519.69
1599.62	8	8	12	7.86	32.24	52.96	41.87	-14.93	-13.87	2	23	12	0.21	0.21	0.13	0.14	1.59	0.32	329.18	481.33	734.29	305.97	3518.7	2519.69
1751.62	8	8	10	7.86	32.24	34.55	41.87	-14.93	-13.87	2	23	12	0.21	0.21	0.16	0.14	1.59	0.48	329.18	481.33	479.03	305.97	3518.7	2519.69
1962.05	8	8	8	7.86	32.24	21.45	41.87	-14.93	-13.87	2	23	12	0.21	0.21	0.21	0.14	1.59	0.83	329.18	481.33	297.36	305.97	3518.7	2519.69
1319.73	14	10	12	33.45	51.94	52.96	13.96	-12.72	-13.87	6	27	12	0.11	0.16	0.13	0.11	1.09	0.32	466.76	660.52	734.29	917.92	4130.65	2519.69
1471.74	14	10	10	33.45	51.94	34.55	13.96	-12.72	-13.87	6	27	12	0.11	0.16	0.16	0.11	1.09	0.48	466.76	660.52	479.03	917.92	4130.65	2519.69
1682.16	14	10	8	33.45	51.94	21.45	13.96	-12.72	-13.87	6	27	12	0.11	0.16	0.21	0.11	1.09	0.83	466.76	660.52	297.36	917.92	4130.65	2519.69
1262.9	14	8	12	33.45	32.24	52.96	13.96	-20.2	-13.87	6	17	12	0.11	0.21	0.13	0.11	1.18	0.32	466.76	651.21	734.29	917.92	2600.78	2519.69
1414.91	14	8	10	33.45	32.24	34.55	13.96	-20.2	-13.87	6	17	12	0.11	0.21	0.16	0.11	1.18	0.48	466.76	651.21	479.03	917.92	2600.78	2519.69
1625.34	14	8	8	33.45	32.24	21.45	13.96	-20.2	-13.87	6	17	12	0.11	0.21	0.21	0.11	1.18	0.83	466.76	651.21	297.36	917.92	2600.78	2519.69
1369.17	12	10	12	19.42	51.94	52.96	13.96	-12.72	-13.87	6	27	12	0.13	0.16	0.13	0.16	1.09	0.32	270.95	660.52	734.29	917.92	4130.65	2519.69
1521.18	12	10	10	19.42	51.94	34.55	13.96	-12.72	-13.87	6	27	12	0.13	0.16	0.16	0.16	1.09	0.48	270.95	660.52	479.03	917.92	4130.65	2519.69
1731.6	12	10	8	19.42	51.94	21.45	13.96	-12.72	-13.87	6	27	12	0.13	0.16	0.21	0.16	1.09	0.83	270.95	660.52	297.36	917.92	4130.65	2519.69
1312.34	12	8	12	19.42	32.24	52.96	13.96	-20.2	-13.87	6	17	12	0.13	0.21	0.13	0.16	1.18	0.32	270.95	651.21	734.29	917.92	2600.78	2519.69
1464.35	12	8	10	19.42	32.24	34.55	13.96	-20.2	-13.87	6	17	12	0.13	0.21	0.16	0.16	1.18	0.48	270.95	651.21	479.03	917.92	2600.78	2519.69
1674.77	12	8	8	19.42	32.24	21.45	13.96	-20.2	-13.87	6	17	12	0.13	0.21	0.21	0.16	1.18	0.83	270.95	651.21	297.36	917.92	2600.78	2519.69
1445.18	10	10	12	12.67	51.94	52.96	13.96	-12.72	-13.87	6	27	12	0.16	0.16	0.13	0.24	1.09	0.32	176.76	660.52	734.29	917.92	4130.65	2519.69
1597.18	10	10	10	12.67	51.94	34.55	13.96	-12.72	-13.87	6	27	12	0.16	0.16	0.16	0.24	1.09	0.48	176.76	660.52	479.03	917.92	4130.65	2519.69
1807.61	10	10	8	12.67	51.94	21.45	13.96	-12.72	-13.87	6	27	12	0.16	0.16	0.21	0.24	1.09	0.83	176.76	660.52	297.36	917.92	4130.65	2519.69
1388.35	10	8	12	12.67	32.24	52.96	13.96	-20.2	-13.87	6	17	12	0.16	0.21	0.13	0.24	1.18	0.32	176.76	651.21	734.29	917.92	2600.78	2519.69
1540.35	10	8	10	12.67	32.24	34.55	13.96	-20.2	-13.87	6	17	12	0.16	0.21	0.16	0.24	1.18	0.48	176.76	651.21	479.03	917.92	2600.78	2519.69
1750.78	10	8	8	12.67	32.24	21.45	13.96	-20.2	-13.87	6	17	12	0.16	0.21	0.21	0.24	1.18	0.83	176.76	651.21	297.36	917.92	2600.78	2519.69
1550.39	8	10	12	7.86	51.94	52.96	13.96	-12.72	-13.87	6	27	12	0.21	0.16	0.13	0.42	1.09	0.32	109.73	660.52	734.29	917.92	4130.65	2519.69
1702.4	8	10	10	7.86	51.94	34.55	13.96	-12.72	-13.87	6	27	12	0.21	0.16	0.16	0.42	1.09	0.48	109.73	660.52	479.03	917.92	4130.65	2519.69
1912.82	8	10	8	7.86	51.94	21.45	13.96	-12.72	-13.87	6	27	12	0.21	0.16	0.21	0.42	1.09	0.83	109.73	660.52	297.36	917.92	4130.65	2519.69
1493.56	8	8	12	7.86	32.24	52.96	13.96	-20.2	-13.87	6	17	12	0.21	0.21	0.13	0.42	1.18	0.32	109.73	651.21	734.29	917.92	2600.78	2519.69
1645.57	8	8	10	7.86	32.24	34.55	13.96	-20.2	-13.87	6	17	12	0.21	0.21	0.16	0.42	1.18	0.48	109.73	651.21	479.03	917.92	2600.78	2519.69
1855.99	8	8	8	7.86	32.24	21.45	13.96	-20.2	-13.87	6	17	12	0.21	0.21	0.21	0.42	1.18	0.83	109.73	651.21	297.36	917.92	2600.78	2519.69
1274.08	14	10	12	33.45	51.94	52.96	13.96	-12.72	-16.64	6	27	10	0.11	0.16	0.13	0.11	1.09	0.27	466.76	660.52	881.15	917.92	4130.65	2099.74
1294.28	14	10	10	33.45	51.94	34.55	13.96	-12.72	-23.77	6	27	7	0.11	0.16	0.16	0.11	1.09	0.28	466.76	660.52	821.19	917.92	4130.65	1469.82

Table 2: Maximum power density = 14.6 W/m

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